

Topic 08: Bernoulli Equations & Composition Substitutions

Professor Leonard Master Series · Lessons 22 to 24.5

Course: Applied Ordinary Differential Equations (ENGR 213 Companion)

1. Whiteboard Intuition: The Bernoulli Transformation

A **Bernoulli Equation** looks almost linear, except for an annoying power of y^n on the right side:

$$\frac{dy}{dx} + P(x)y = Q(x)y^n$$

Professor Leonard's 4-Step Whiteboard Recipe:

1. **Divide every single term by y^n :**

$$y^{-n} \frac{dy}{dx} + P(x)y^{1-n} = Q(x)$$

2. **Notice the magic substitution:** Let $w = y^{1-n}$.
3. **Differentiate using chain rule:**

$$\frac{dw}{dx} = (1-n)y^{-n} \frac{dy}{dx} \implies y^{-n} \frac{dy}{dx} = \frac{1}{1-n} \frac{dw}{dx}$$

4. **Plug back in:**

$$\frac{1}{1-n} \frac{dw}{dx} + P(x)w = Q(x) \implies \frac{dw}{dx} + (1-n)P(x)w = (1-n)Q(x)$$

It has turned into a standard First-Order Linear ODE! Solve for w , then back-substitute $w = y^{1-n}$.

2. Composition Substitutions: $\frac{dy}{dx} = f(Ax + By + C)$

When you see an expression like $\frac{dy}{dx} = \tan^2(x + y)$ or $\frac{dy}{dx} = (2x + 3y + 1)^2$:

- Let $u = Ax + By + C$.
- Take derivative: $\frac{du}{dx} = A + B \frac{dy}{dx} \implies \frac{dy}{dx} = \frac{1}{B} \left(\frac{du}{dx} - A \right)$.

- Substitute to get:

$$\frac{1}{B} \left(\frac{du}{dx} - A \right) = f(u) \implies \frac{du}{dx} = A + Bf(u)$$

This is instantly separable:

$$\frac{du}{A + Bf(u)} = dx$$